

Lecture 1: Course Overview

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- └ DBMS Fundamentals
 - └ Purpose of Databases
 - └ Definition
 - └ Manages enterprise information
 - └ Handles collection of interrelated data
 - └ Provides programs to access and manipulate data
 - └ Characteristics
 - └ Typically very large in size
 - └ Provides a convenient and efficient environment
 - └ Modern Applications
 - └ Financial Services
 - └ Banking (Net Banking)
 - └ Travel & Hospitality
 - └ Reservations (Airline, Railway, Hotel)
 - └ Education
 - └ University Systems (students, courses, grades)
 - └ E-Commerce
 - └ Online Retail (Amazon, Flipkart, E-bay)
 - └ Other Sectors
 - └ Manufacturing, Human Resources, Social Media
- └ File Systems vs. DBMS
 - └ Drawbacks of File Systems
 - └ Data Redundancy & Inconsistency
 - └ Same data stored in multiple places, leading to conflicts
 - └ Difficulty in Accessing Data
 - └ Non-trivial task to access data from multiple files
 - └ Data Isolation
 - └ Data scattered in various files and formats
 - └ Integrity Problems
 - └ Hard to add new constraints (e.g., positive balance)
 - └ Atomicity Issues
 - └ Failures can leave database in an inconsistent state (e.g., fund transfer)
 - └ Concurrent Access Anomalies

- └─ Uncontrolled concurrent access leads to inconsistencies
 - └─ Security Problems
 - └─ Hard to provide user-level security
- └─ Course Logistics & Roadmap
 - └─ Course Structure
 - └─ Format
 - └─ 40 modules, approx. 30 minutes each
 - └─ 8-week duration
 - └─ Content Split
 - └─ Weeks 1-5: Application Programming
 - └─ Weeks 5-8: System Analysis & Design
 - └─ Learning Goals (Career Paths)
 - └─ Application Programmer
 - └─ Focus on querying and data manipulation
 - └─ Database Analyst
 - └─ Focus on design, performance tuning, indexing
 - └─ DBMS Designer/Engineer
 - └─ Focus on creating database management systems
 - └─ Resources
 - └─ Textbook
 - └─ Database System Concepts by Silberschatz, Korth, Sudarshan (6th Ed.)
 - └─ Teaching Assistants (TAs)
 - └─ Srijoni Majumdar
 - └─ Himadri Bhushan Bhuyan
 - └─ Gurunath Reddy
- └─ Prerequisites & Preparation
 - └─ Essential Topics
 - └─ Discrete Mathematics
 - └─ Set Theory (Union, Intersection, etc.)
 - └─ Relations & Functions (Reflexive, Symmetric, etc.)
 - └─ Logic (Propositional and Predicate)
 - └─ Computer Science Fundamentals
 - └─ Data Structures (BST, B-Trees, Hashing)
 - └─ Algorithms (Sorting, Searching)
 - └─ Programming in C
 - └─ Desirable Topics
 - └─ Object-Oriented Concepts

- | |— 00 Analysis and Design
- | |— Programming in C++ or Java
- └ Self-Assessment
 - └ Assignment Zero
 - |— Covers prerequisite knowledge
 - └ Not graded; for self-evaluation only